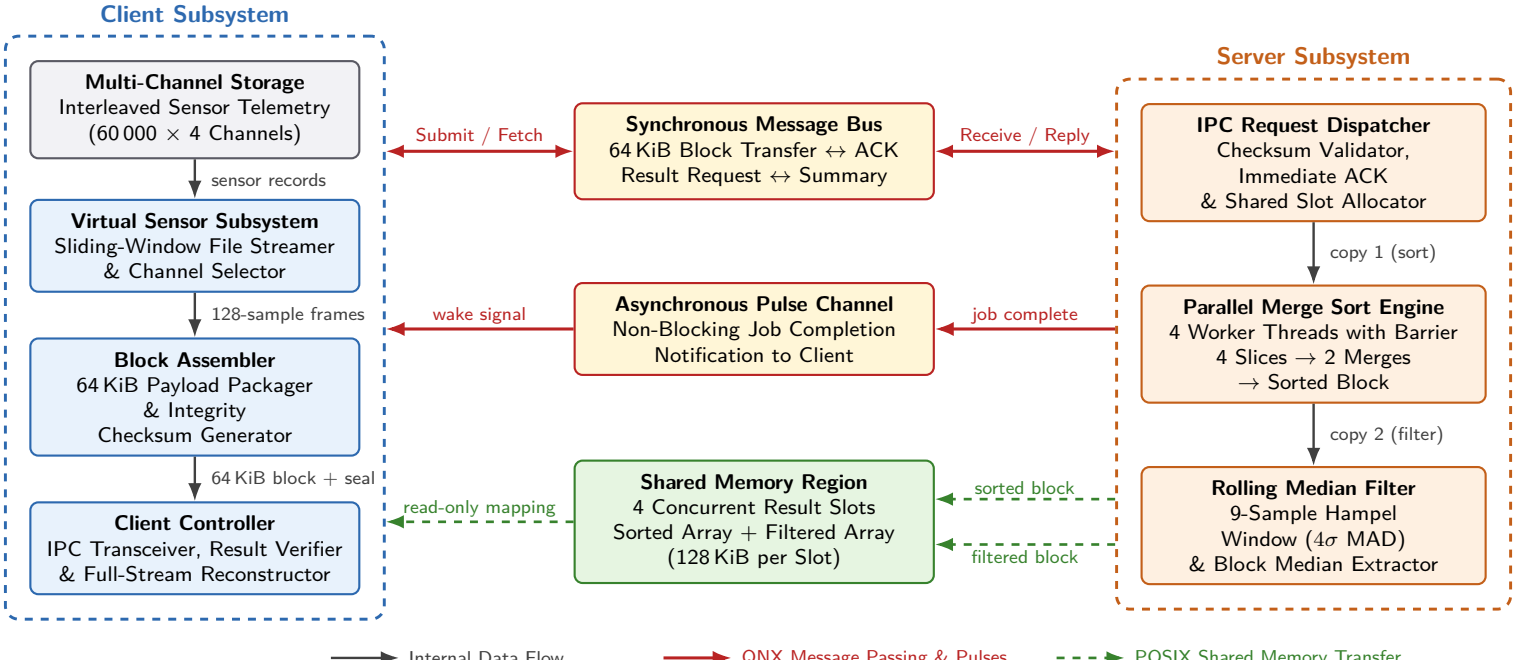

Client Subsystem
Multi-Channel Storage
Interleaved Sensor Telemetry
(60 000 × 4 Channels)
sensor records
Virtual Sensor Subsystem
Sliding-Window File Streamer
& Channel Selector
128-sample frames
Block Assembler
64 KiB Payload Packager
& Integrity
Checksum Generator
64 KiB block + seal
Client Controller
IPC Transceiver, Result Verifier
& Full-Stream Reconstructor
Submit / Fetch
Synchronous Message Bus
64 KiB Block Transfer ↔ ACK
Result Request ↔ Summary
Receive / Reply
wake signal
Asynchronous Pulse Channel
Non-Blocking Job Completion
Notification to Client
job complete
read-only mapping
Shared Memory Region
4 Concurrent Result Slots
Sorted Array + Filtered Array
(128 KiB per Slot)
sorted block
filtered block
Server Subsystem
IPC Request Dispatcher
Checksum Validator,
Immediate ACK
& Shared Slot Allocator
copy 1 (sort)
Parallel Merge Sort Engine
4 Worker Threads with Barrier
4 Slices → 2 Merges
→ Sorted Block
copy 2 (filter)
Rolling Median Filter
9-Sample Hampel
Window (4σ MAD)
& Block Median Extractor
Internal Data Flow
QNX Message Passing & Pulses
POSIX Shared Memory Transfer